Fundamental Trigonometric Identities

ANSWER KEY

Using the Pythagorean identities

1 Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta \;= -\frac{4}{5}$ **b** $\tan\theta \;= -\frac{3}{4}$ **c** $\sec\theta \;= -\frac{5}{4}$ **d** $\csc\theta \;= \frac{5}{3}$

2 Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.
$\sec^2\theta = 1 + \tan^2\theta = 5$, and in QIII $\cos\theta < 0$, so $\cos\theta = -\frac{\sqrt{5}}{5}$ and $\sin\theta = \tan\theta\cos\theta = -\frac{2\sqrt{5}}{5}$.

Simplifying and verifying

3 Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x} \;= \frac{\sin^2 x}{\sin x} = \sin x$

b $\sec x \cot x \;= \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$

c $\tan x \csc x \;= \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

4 Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.
$(1+\tan^2 x)\cos^2 x = \sec^2 x \cos^2 x = \frac{1}{\cos^2 x}\cos^2 x = 1$.

5 Verify the identity $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$.
Cross-multiplying view: $\sin^2 x = (1+\cos x)(1-\cos x) = 1-\cos^2 x$, which is the Pythagorean identity.
Equivalently, multiply the left side by $\frac{1-\cos x}{1-\cos x}$ to obtain $\frac{\sin x(1-\cos x)}{1-\cos^2 x} = \frac{\sin x(1-\cos x)}{\sin^2 x} = \frac{1-\cos x}{\sin x}$.